Advanced Control Systems

Assignment 2: Kinetic Energy, Potential Energy, and Generalized Inertia Matrix $\mathrm{B}(q)$ Derivation for a 3-DOF PRR Manipulator

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Abstract

This report presents the derivation of the total kinetic energy $\mathcal{T}(q, \dot{q})$, gravitational potential energy $\mathcal{U}(q)$, and generalized inertia matrix $\mathbf{B}(q)$ for a spatial 3-DOF PRR (Prismatic-Revolute-Revolute) robotic manipulator. Using König's theorem, partial Jacobians associated with link centers of mass (CoM), and local inertia tensors, closed-form parametric equations are established. The physical parameters are evaluated using exact rational symbolic arithmetic to prevent numerical round-off artifacts. Furthermore, numerical evaluation and 3D manipulator visualization are conducted at a sample configuration strictly respecting the URDF joint limits ($q_1 \in [-0.5, 0.0]$ m, $q_2, q_3 \in [-\pi, \pi]$ rad).

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1 System Architecture & Mass Properties

1.1 Kinematic and Dynamic Parameter Specifications

What is being done: The physical parameters extracted from the URDF file for each body link are defined as follows:

1. **Link 1 (Prismatic Joint, q_1):** Mass $m_1 = 1.0$ kg, prismatic joint limit $q_1 \in [-0.5, 0.0]$ m. Geometry: rectangular bar (0.036 m $\times$ 0.036 m $\times$ 0.5 m).
2. **Link 2 (Revolute Joint, q_2):** Mass $m_2 = 1.0$ kg, revolute joint limit $q_2 \in [-\pi, \pi]$ rad. Geometry: solid cylinder along Z_2 ($r = 0.019$ m, $h = 0.4$ m).
3. **Link 3 (Revolute Joint, q_3):** Mass $m_3 = 1.0$ kg, revolute joint limit $q_3 \in [-\pi, \pi]$ rad. Geometry: solid cylinder along X_3 ($r = 0.019$ m, $h = 0.4$ m).

Why it is done: Mass and inertia distribution dictate the dynamic response of the manipulator. Establishing accurate geometric inertia tensors relative to each link's Center of Mass (CoM) is required prior to applying Lagrangian energy principles.

2 Energy Derivation via Lagrangian Mechanics

2.1 Kinetic Energy $\mathcal{T}(q, \dot{q})$ Formulation

What is being done: According to König's theorem, the total kinetic energy $\mathcal{T}(q, \dot{q})$ of an n-link manipulator is the sum of translational and rotational kinetic energies of all individual links:

$$\mathcal{T}(q, \dot{q}) = \sum_{i=1}^{3} \left(\frac{1}{2} m_i \dot{\mathbf{p}}_{l_i}^T \dot{\mathbf{p}}_{l_i} + \frac{1}{2} \boldsymbol{\omega}_i^T \mathbf{I}_{l_i} \boldsymbol{\omega}_i \right) = \frac{1}{2} \dot{q}^T \mathbf{B}(q) \dot{q} \tag{1}$$

Expressing CoM linear velocity $\dot{\mathbf{p}}_{l_i} = \mathbf{J}_P^{l_i}(q)\dot{q}$ and angular velocity $\boldsymbol{\omega}_i = \mathbf{J}_O^{l_i}(q)\dot{q}$ via partial Jacobians yields the generalized inertia matrix $\mathbf{B}(q)$:

$$\mathbf{B}(q) = \sum_{i=1}^{3} \left(m_i (\mathbf{J}_P^{l_i})^T \mathbf{J}_P^{l_i} + (\mathbf{J}_O^{l_i})^T \mathbf{R}_i \mathbf{I}_{l_i}^i \mathbf{R}_i^T \mathbf{J}_O^{l_i} \right) \tag{2}$$

Why it is done: The generalized inertia matrix $\mathbf{B}(q) \in \mathbb{R}^{3\times 3}$ is symmetric and positive-definite ($\mathbf{B}(q) = \mathbf{B}(q)^T > 0$). It maps joint velocity vectors to generalized kinetic energy and constitutes the principal term in the manipulator dynamics.

2.2 Potential Energy $\mathcal{U}(q)$ Formulation

What is being done: The gravitational potential energy $\mathcal{U}(q)$ is calculated as the sum of potential energies of the link centers of mass relative to the base frame Σ_0:

$$\mathcal{U}(q) = -\sum_{i=1}^{3} m_i \mathbf{g}_0^T \mathbf{p}_{l_i}(q) \tag{3}$$

where $\mathbf{g}_0 = [0, 0, -g]^T$ with $g = 9.81$ m/s^2.

Substituting the vertical CoM positions $z_{Ci}(q)$ yields:

$$\mathcal{U}(q) = g\left[m_1(q_1 + 0.40) + m_2(q_1 + 1.05) + m_3\left(q_1 + 1.05 + 0.20\sin q_3\right)\right] \tag{4}$$

Evaluating with link masses $m_i = 1$ kg:

$$\mathcal{U}(q) = 9.81 \cdot \left[3q_1 + 2.50 + 0.20\sin q_3\right] \text{ J} \tag{5}$$

Why it is done: $\mathcal{U}(q)$ depends solely on joint position coordinates q. Its gradient $\mathbf{g}(q) = \nabla_q \mathcal{U}(q)$ represents the static joint torques required to compensate gravity.

3 Symbolic and Numerical Matrix Expressions

3.1 Closed-Form Inertia Matrix $\mathrm{B}(q)$

Evaluating the symbolic terms yields:

$$\mathbf{B}(q) = \begin{bmatrix} m_1 + m_2 + m_3 & 0 & \frac{m_3}{5}\cos q_3 \\ 0 & I_{2yy} + I_{3yy}\cos^2 q_3 + I_{3xx}\sin^2 q_3 + \frac{m_3}{25}\cos^2 q_3 & 0 \\ \frac{m_3}{5}\cos q_3 & 0 & I_{3zz} + \frac{m_3}{25} \end{bmatrix} \tag{6}$$

Substituting exact physical parameters yields the evaluated numerical matrix:

$$\mathbf{B}(q) = \begin{bmatrix} 3.0 & 0 & 0.2\cos q_3 \\ 0 & 0.1068 - 0.05324\sin^2 q_3 & 0 \\ 0.2\cos q_3 & 0 & 0.05342 \end{bmatrix} \tag{7}$$

4 Sample Numerical Evaluation & 3D Visualization

4.1 Sample State inside URDF Joint Limits

What is being done: To verify the physical validity of the energy model, we evaluate $\mathbf{B}(q)$, $\mathcal{T}(q, \dot{q})$, and $\mathcal{U}(q)$ at a specific configuration q_{sample} strictly within URDF limits:

$$q_1 = -0.25 \text{ m} \quad (\text{URDF Limit: } [-0.5, 0.0] \text{ m}) \tag{8}$$

$$q_2 = \frac{\pi}{4} \text{ rad} \approx 0.7854 \text{ rad} \quad (\text{URDF Limit: } [-\pi, \pi] \text{ rad}) \tag{9}$$

$$q_3 = \frac{\pi}{6} \text{ rad} \approx 0.5236 \text{ rad} \quad (\text{URDF Limit: } [-\pi, \pi] \text{ rad}) \tag{10}$$

Joint velocity vector: $\dot{q}_{\text{sample}} = [0.11 \text{ m/s}, 0.13 \text{ rad/s}, -0.27 \text{ rad/s}]^T$.

Numerical Results: Evaluating at state $(q_{\text{sample}}, \dot{q}_{\text{sample}})$ yields:

$$\mathbf{B}(q_{\text{sample}}) = \begin{bmatrix} 3.0000 & 0 & 0.1732 \\ 0 & 0.0535 & 0 \\ 0.1732 & 0 & 0.0534 \end{bmatrix} \tag{11}$$

$$\mathcal{T}(q_{\text{sample}}, \dot{q}_{\text{sample}}) = 0.015406 \text{ J} \tag{12}$$

$$\mathcal{U}(q_{\text{sample}}) = 18.148500 \text{ J} \tag{13}$$

$$\mathcal{E}_{\text{total}} = \mathcal{T} + \mathcal{U} = 18.163906 \text{ J} \tag{14}$$

Why it is done: This sample state provides concrete numerical validation of the mathematical energy equations and serves as a benchmark for controller verification.

4.2 3D Robot Configuration Visualization

Figure 1 shows the 3D plot of the 3-DOF PRR manipulator generated in MATLAB at the selected sample configuration $q_{\text{sample}} = [-0.25 \text{ m}, 0.7854 \text{ rad}, 0.5236 \text{ rad}]^T$.

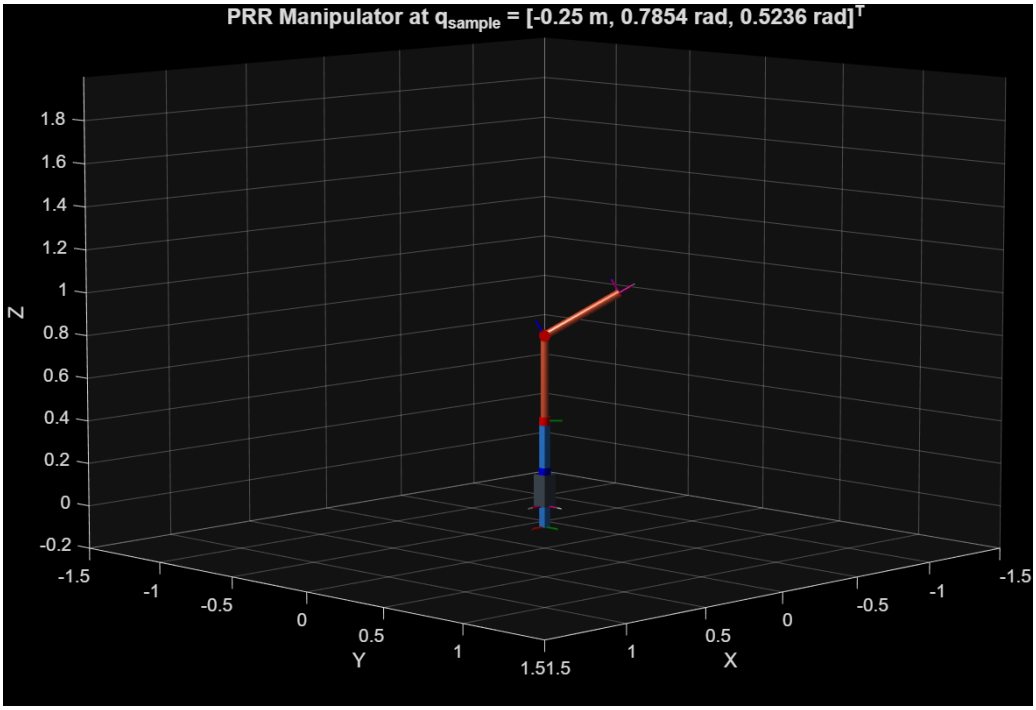


Figure 1: 3D Visualization at $q_{\text{sample}} = [-0.25 \text{ m}, 0.7854 \text{ rad}, 0.5236 \text{ rad}]^T$ inside joint limits.